■ B.7.6 External Functions

If you include the appropriate hooks in an external program, then you can call individual functions in the external program directly from *Mathematica*. In addition, you can have the external program call *Mathematica* back to perform calculations. All the necessary communications are done through a bidirectional pipe between *Mathematica* and the external program.

Mathematica sends "packets" containing requests for particular external function evaluations; the evaluations are performed by the external program, and the results are sent as a packet back to *Mathematica*. Routines for decoding and creating packets in the external program are typically distributed in source code form with *Mathematica*.

`MathInit()`	initialize communications
`MathInstall(`*f*`, "`*fname*`", "`*ftype*`", "`*argnames*`", "`*argtypes*`")`	install the C function with pointer *f* so that it can be called from *Mathematica* with the name *fname*
`MathStart()`	put the external program in a state to be called by *Mathematica*
`MathExec("`*expr*`")`	evaluate *expr* as *Mathematica* input, and return the result as a string

Typical C functions for setting up communication with *Mathematica*.

```
main()
{
int BitAnd();
double DoubleShift();

MathInit();

MathInstall(BitAnd, "BitAnd", "int", "x_Integer, y_Integer", "int, int");

MathInstall(DoubleShift, "DoubleShift", "double", "x, n_Integer", "double, int");

MathExec("BitAnd::usage = \"BitAnd[x, y] computes the bitwise AND of x and y.\"");

MathExec(
"DoubleShift::usage = \"DoubleShift[x] shifts the double representation of x by n bits.\"");

MathStart();
}
```

Web sample page from The *Mathematica* Book, First Edition, by Stephen Wolfram, published by Addison-Wesley Publishing Company (hardcover ISBN 0-201-19334-5; softcover ISBN 0-201-19330-2). To order *Mathematica* or this book contact Wolfram Research: info@wolfram.com; http://www.wolfram.com/; 1-800-441-6284.

```
int BitAnd(x, y)
int x, y;
{
return( x&y );
}

double DoubleShift(x, n)
double x;
int n;
{
return( x >> n );
}
```

An external C program set up to be called from *Mathematica*.

The C types typically handled by `MathInstall` are `int`, `double` and `char *`.